KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Definite and Logarithmic Substitution

Name _____

S1

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{3(6x+1)}{3x^2+x+5}\,dx$$

S2

Evaluate each definite integral using substitution.

$$\int_0^2 6(3x+1)^3\,dx$$

S3

Evaluate each definite integral using substitution.

$$\int_0^1 3(2x+5)(x^2+5x+2)^3\,dx$$

S4

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{2(4x-3)}{2x^2-3x+7}\,dx$$

KAIZEN MATHS

Lesson starter

Integration by Substitution and Parts

Definite and Logarithmic Substitution

Name _____

S1

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{3(6x+1)}{3x^2+x+5}\,dx$$

S2

Evaluate each definite integral using substitution.

$$\int_0^2 6(3x+1)^3\,dx$$

S3

Evaluate each definite integral using substitution.

$$\int_0^1 3(2x+5)(x^2+5x+2)^3\,dx$$

S4

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{2(4x-3)}{2x^2-3x+7}\,dx$$

ALG-99DB-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Definite and Logarithmic Substitution

Name _____

EA1

Evaluate each definite integral using substitution.

$$\int_2^5 4(2x+2)^4\,dx$$

EA2

Evaluate each definite integral using substitution.

$$\int_1^2 2(2x+1)(x^2+x+2)^3\,dx$$

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Definite and Logarithmic Substitution

Name _____

EB1

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{3(4x+5)}{2x^2+5x+6}\,dx$$

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Definite and Logarithmic Substitution

Name _____

EA1

Evaluate each definite integral using substitution.

$$\int_2^5 4(2x+2)^4\,dx$$

EA2

Evaluate each definite integral using substitution.

$$\int_1^2 2(2x+1)(x^2+x+2)^3\,dx$$

ALG-99DB-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

Integration by Substitution and Parts

Definite and Logarithmic Substitution

Name _____

EB1

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{3(4x+5)}{2x^2+5x+6}\,dx$$

ALG-99DB-L2 · EB

Show your working

Teacher answers & working

Integration by Substitution and Parts · Definite and Logarithmic Substitution

ALG-99DB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{3(6x+1)}{3x^2+x+5}\,dx$$

Answer:

$3\ln|3x^2+x+5|+C$

Working:

Let $u = 3x^2+x+5$.

Then $\frac{du}{dx} = 6x+1$, so $du = (6x+1)\,dx$.

The numerator contains $3(6x+1)$, so the integral becomes $\int \frac{3}{u}\,du$.

Use $\int \frac{1}{u}\,du = \ln|u|+C$.

Therefore the integral is $3\ln|u|+C$.

Replace u with $3x^2+x+5$: $3\ln|3x^2+x+5|+C$.

S2

Evaluate each definite integral using substitution.

$$\int_0^2 6(3x+1)^3\,dx$$

Answer:

1200

Working:

Let $u = 3x+1$, so $du = 3\,dx$ and $dx = \frac{1}{3}\,du$.

Change the limits: when $x = 0$, $u = 3(0)+1 = 1$.

When $x = 2$, $u = 3(2)+1 = 7$.

The integral becomes $\int_1^7 2u^3\,du$.

Integrate: $\left[\frac{1}{2}u^4\right]_1^7$.

Upper value: $F(7) = \frac{1}{2}(7)^4$.

Lower value: $F(1) = \frac{1}{2}(1)^4$.

Subtract: $\frac{1}{2}[(7)^4-(1)^4]$.

Evaluate the powers: $\frac{1}{2}[(2401)-(1)]$.

1200.

Teacher answers & working

Integration by Substitution and Parts · Definite and Logarithmic Substitution

ALG-99DB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S3

Evaluate each definite integral using substitution.

$$\int_0^1 3(2x+5)(x^2+5x+2)^3\,dx$$

Answer:

3060

Working:

Let $u = x^2 + 5x + 2$.

Then $du = (2x+5)\,dx$.

Lower limit: $u = 1(0)^2 + (5)(0) + (2) = 2$.

Upper limit: $u = 1(1)^2 + (5)(1) + (2) = 8$.

The integral becomes $\int_2^8 3u^3\,du$.

Integrate: $\left[\frac{3}{4}u^4\right]_2^8$.

Upper value: $F(8) = \frac{3}{4}(8)^4$.

Lower value: $F(2) = \frac{3}{4}(2)^4$.

Subtract: $\frac{3}{4}[(8)^4 - (2)^4]$.

Evaluate the powers: $\frac{3}{4}[(4096) - (16)]$.

3060.

S4

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{2(4x-3)}{2x^2-3x+7}\,dx$$

Answer:

$2\ln|2x^2 - 3x + 7| + C$

Working:

Let $u = 2x^2 - 3x + 7$.

Then $\frac{du}{dx} = 4x - 3$, so $du = (4x-3)\,dx$.

The numerator contains $2(4x-3)$, so the integral becomes $\int \frac{2}{u}\,du$.

Use $\int \frac{1}{u}\,du = \ln|u| + C$.

Therefore the integral is $2\ln|u| + C$.

Replace u with $2x^2 - 3x + 7$: $2\ln|2x^2 - 3x + 7| + C$.

Teacher answers & working

Integration by Substitution and Parts · Definite and Logarithmic Substitution

ALG-99DB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Evaluate each definite integral using substitution.

$$\int_{2}^{5} 4(2x+2)^{4}\,dx$$

Answer:

$$\frac{482112}{5}$$

Working:

Let $u = 2x + 2$, so $du = 2\,dx$ and $dx = \frac{1}{2}\,du$.

Change the limits: when $x = 2$, $u = 2(2) + 2 = 6$.

When $x = 5$, $u = 2(5) + 2 = 12$.

The integral becomes $\int_{6}^{12} 2u^{4}\,du$.

Integrate: $\left[\frac{2}{5}u^{5}\right]_{6}^{12}$.

Upper value: $F(12) = \frac{2}{5}(12)^{5}$.

Lower value: $F(6) = \frac{2}{5}(6)^{5}$.

Subtract: $\frac{2}{5}[(12)^{5} - (6)^{5}]$.

Evaluate the powers: $\frac{2}{5}[(248832) - (7776)]$.

$\frac{482112}{5}$.

Teacher answers & working

Integration by Substitution and Parts · Definite and Logarithmic Substitution

ALG-99DB-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA2

Evaluate each definite integral using substitution.

$$\int_1^2 2(2x+1)(x^2+x+2)^3\,dx$$

Answer:

1920

Working:

Let $u = x^2 + x + 2$.

Then $du = (2x+1)\,dx$.

Lower limit: $u = 1(1)^2 + (1)(1) + (2) = 4$.

Upper limit: $u = 1(2)^2 + (1)(2) + (2) = 8$.

The integral becomes $\int_4^8 2u^3\,du$.

Integrate: $\left[\frac{1}{2}u^4\right]_4^8$.

Upper value: $F(8) = \frac{1}{2}(8)^4$.

Lower value: $F(4) = \frac{1}{2}(4)^4$.

Subtract: $\frac{1}{2}[(8)^4 - (4)^4]$.

Evaluate the powers: $\frac{1}{2}[(4096) - (256)]$.

1920.

EB1

Integrate each expression by recognising a derivative-over-function form.

$$\int \frac{3(4x+5)}{2x^2+5x+6}\,dx$$

Answer:

$3\ln|2x^2+5x+6| + C$

Working:

Let $u = 2x^2 + 5x + 6$.

Then $\frac{du}{dx} = 4x + 5$, so $du = (4x+5)\,dx$.

The numerator contains $3(4x+5)$, so the integral becomes $\int \frac{3}{u}\,du$.

Use $\int \frac{1}{u}\,du = \ln|u| + C$.

Therefore the integral is $3\ln|u| + C$.

Replace u with $2x^2+5x+6$: $3\ln|2x^2+5x+6| + C$.